Online Appendix to:
Frequent Losers: A Cover-Bidder Screen Without Bid Microdata

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Abstract

This online appendix accompanies *Frequent Losers: A Participation-Based Screen for Public Procurement* and contains: (A) the cover-bidding framework and proofs of the three formal results invoked in the main text; (B) core supporting tables and figures; (C) identification audits (Oster bounds and the anti-leakage audit); (D) threshold and specification robustness tables; (E) operational metrics and detector comparison; and (F) the staggered-design estimators that return null at the available bandwidths. References to sections, equations, and tables in the main text use the same labels as the published manuscript.

Appendix A. Cover-Bidding Framework and Proofs

This appendix formalises the separating-equilibrium framework the paper reads against. A cartel of n firms in a procurement market controls a designated winner with bid b^* and deploys $m \geq 0$ cover bidders, each submitting a bid $b_\ell > b^*$. Per-bidder deployment cost is $c_1 > 0$, and the cartel faces a detection probability $\theta_k \in (0, 1)$ that depends on oversight capacity in market k—smaller, less-supervised buyers have lower θ_k. The cartel maximises expected surplus $\pi(m, \theta_k, c_1)$, the difference between cartel rents and the sum of cover-bidder costs and expected detection penalties, under two standard assumptions: A1 (profit positivity, $\partial\pi/\partial m\big|_{m=0} > 0$) and A2 (diminishing returns, $\partial^2\pi/\partial m^2 < 0$). The empirical environment we read it against is described by

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award-record data: each firm i has a participation count tenders_count_i, a win indicator $\text{wins}_i \in \{0, 1\}$, and a tenure τ_i. The framework delivers four results. Lemma 1 identifies the wins $= 0$ filter as the equilibrium choice of the cover-bidder type. Proposition 2 identifies $\log(1 + \text{tenders_count})$ as a sufficient ranking statistic for cover-bidder type given award-record data. Proposition 3 characterises the cartel's optimal deployment rate m^* in the principal cost of detection θ_k. And Proposition 4 formalises why the broad-sample association β and the overlap-restricted association β^{ov} are different empirical objects under the framework's deployment problem.

Lemma 1 (Wins-zero separating equilibrium). *Suppose cover bidders draw their bid b_C from a strategy that satisfies (a) $b_C > b^*$ in equilibrium (cartel-allocation constraint) and (b) winning triggers either capacity revelation (undermining the cover identity in subsequent rounds) or internal cartel sanctions on the deviating firm. Genuine entrants of type G face no such constraint and accept any $b_G < b^*$ that yields positive expected payoff. Then in the separating equilibrium, cover bidders satisfy $\Pr(\textit{win} \mid C) = 0$: the bright-line filter wins $= 0$ is the equilibrium choice of type C.*

Proof of Lemma 1. Write the cover-bidder per-period payoff as $u_C(b) = -c_1 - \theta_k \phi_0 \cdot \mathbf{1}[\text{win}] - \kappa \cdot \mathbf{1}[\text{win}]$, where $\phi_0 > 0$ is the marginal expected detection penalty per cover bid and $\kappa > 0$ captures expected cartel-internal sanction or capacity-revelation cost. Choosing $b_C > b^*$ delivers $\Pr(\text{win} \mid b_C, b^*) = 0$ in any equilibrium where the designated winner submits b^*, so $u_C = -c_1$. Any deviation to $b'_C < b^*$ delivers $u'_C = -c_1 - \theta_k \phi_0 - \kappa + \text{rents}$, where rents is the deviation gain from winning. Under the cartel's allocation rule rents $< \kappa$ for any cover bidder (otherwise the allocation would not be incentive-compatible), so $u_C > u'_C$ and the equilibrium strategy is $b_C > b^*$ with $\Pr(\text{win} \mid C) = 0$. Type G has no κ penalty and accepts $b_G < b^*$ when expected payoff is positive; the empirical filter wins $= 0$ identifies the type-C subset, conditional on the cartel deploying its allocation rule. □ □

Proposition 2 (Linearised cumulative exposure). *Suppose firm i is a cover bidder of type C deployed by a cartel in market k at per-period rate $\lambda_C(\theta_k, c_1)$ and tenure τ_i. Under A1–A2 and a stationary deployment process, $\mathbb{E}[\textit{tenders_count}_i \mid \textit{type} = C, \tau_i] = \lambda_C(\theta_k, c_1)\, \tau_i$. For genuine competitive entrants of type G, $\mathbb{E}[\textit{tenders_count}_i \mid \textit{type} = G, \tau_i] = \lambda_G\, \tau_i$ with $\lambda_G < \lambda_C$ in any cartel-targeted market. The log-statistic $\log(1 + \textit{tenders_count}_i)$ is monotone in λ given tenure, and—under monotone likelihood*

ratio in the Poisson approximation $T_i \sim \text{Poisson}(\lambda\tau_i)$*—the posterior* $\Pr(\textit{type} = C \mid \textit{tenders_count}_i = t)$ *is non-decreasing in* t*. Combined with Lemma 1, the joint statistic* $(\log(1 + \textit{tenders_count}), \mathbf{1}[\textit{wins} = 0])$ *is a sufficient screen for the cover-bidder type given award-record data; the binary frequent-loser rule is the information-coarsening of this latent posterior under the separating equilibrium.*

Proof of Proposition 2. The expectation result is immediate from the law of large numbers applied to the deployment counting process given constant rate λ. The monotonicity claim follows from Bayes' rule: $\Pr(C \mid t) = \Pr(t \mid C)\,\pi_C / [\Pr(t \mid C)\,\pi_C + \Pr(t \mid G)\,(1 - \pi_C)]$ is non-decreasing in t if and only if the likelihood ratio $\Pr(t \mid C)/\Pr(t \mid G)$ is non-decreasing in t. For Poisson likelihoods with rates $\lambda_C > \lambda_G$, this monotone-likelihood-ratio property is standard. Hence $\log(1 + \text{tenders_count})$ is a sufficient ranking statistic for the cover-bidder type and the binary indicator $\mathbf{1}[\text{tenders_count} > k]$ is its information-coarsening form. □ □

Proposition 3 (Optimal cover-bidder deployment). *Under A1–A2, the cartel's optimal cover-bidder deployment rate* m^* *satisfies the first-order condition* $\partial\pi/\partial m = c_1 + \theta_k\phi_0$*, where* $\phi_0 > 0$ *is the marginal expected detection penalty per cover bidder evaluated at the optimum. The implicit comparative statics* $\partial m^*/\partial\theta_k = -\phi_0/(\partial^2\pi/\partial m^2) < 0$ *and* $\partial m^*/\partial c_1 = -1/(\partial^2\pi/\partial m^2) < 0$ *follow from A2 (concavity ensures the denominator is negative).*

Proof of Proposition 3. The first-order condition follows from differentiating $\pi(m, \theta_k, c_1)$ with respect to m and equating the marginal cartel surplus to the marginal cost of the cover bidder. By A2 the implicit interior solution defined by the FOC is unique, and implicit differentiation of the FOC yields the displayed comparative statics. The empirical implication is that cover-bidder deployment is more aggressive where the procuring unit's principal cost of detection is lower (§7.3). □ □

Propositions 2–3 together imply that frequent-loser participation is not distributed uniformly over items: under A1–A2, the cartel deploys cover bidders preferentially in items where its expected surplus is highest, conditional on the deployment cost c_1 and the local detection probability θ_k. Items the cartel selects into therefore differ from items it does not on a vector of unobserved characteristics $\mathbf{u}$ that governs the cartel's expected surplus—chiefly, the rents available from the designated winner's bid relative to the

competitive equilibrium price. The next proposition formalises the implication for the two empirical objects β and β^{ov}.

Proposition 4 (Screening-value vs. treatment-effect objects). *Let $D_i \in \{0, 1\}$ indicate frequent-loser presence in item i, $\mathbf{x}_i$ the vector of observables absorbed by fixed effects and matching covariates, and $\mathbf{u}_i$ the vector of unobservables governing the cartel's deployment optimum. Suppose the cartel's optimal deployment satisfies $\Pr(D_i = 1 \mid \mathbf{x}_i, \mathbf{u}_i) = g(\mathbf{x}_i, \mathbf{u}_i)$ with g strictly increasing in the rent component of $\mathbf{u}_i$. Then (1) the broad-sample $\beta = \mathbb{E}[y_i \mid D_i = 1, \mathbf{x}_i] - \mathbb{E}[y_i \mid D_i = 0, \mathbf{x}_i]$ integrates over conditional distributions of $\mathbf{u}_i$ that differ systematically in the rent component; (2) the overlap-restricted β^{ov} within the common-support cell Ω trims attention to items where the rent component takes intermediate values, so the conditional distributions of $\mathbf{u}_i$ across $D_i = 1$ and $D_i = 0$ within Ω are more comparable; and (3) if the rent component shifts both the deployment probability and the realised price in the same direction, $\beta > 0$ on the broad sample together with $\beta^{ov} < 0$ is the prediction of the deployment problem, not its refutation: forcing overlap removes precisely the items where the participation footprint and the price imprint co-occur.*

Proof of Proposition 4. Claim 1 follows from the law of iterated expectations: the broad-sample β averages $\mathbb{E}[y_i \mid D_i, \mathbf{x}_i, \mathbf{u}_i]$ over the conditional distributions of $\mathbf{u}_i$ given D_i and $\mathbf{x}_i$, and Proposition 3 implies $\mathbf{u}_i \mid D_i = 1$ has a higher rent component than $\mathbf{u}_i \mid D_i = 0$ at the same $\mathbf{x}_i$. Claim 2 follows from the definition of common support: cells in Ω have propensity score $\Pr(D_i = 1 \mid \mathbf{x}_i)$ bounded away from 0 and 1, which by strict monotonicity of g in the rent component implies that the rent component takes intermediate values within Ω. Claim 3 is the comparative statement: averaging over $\mathbf{u}_i$ on the broad sample produces a positive β while restricting to Ω trims the high-rent cells out of the average, leaving a β^{ov} that need not be positive and may be negative for reasons unrelated to the treatment itself. □ □

The proposition is the formal counterpart to the screening-value reading in §7.2: under Proposition 4's rent-component-co-movement premise, the sign reversal is the model's prediction. The model does not by itself adjudicate whether the premise holds; the empirical evidence of §6, §8.1, and the segment-level decomposition of §7.2 together discipline the interpretation.

The framework is silent on the bid-level shape of b_C beyond the equilibrium constraint $b_C > b^*$. The early empirical cover-bidding literature (Porter and Zona, 1993, 1999;

Bajari and Ye, 2003) reads the constraint loosely: cover bidders submit bids *visibly* above the winning bid because the cartel's purpose is to manufacture the appearance of competition, not to compete credibly (the *textbook* reading, R1). A more recent strand reads it tighter: cover bids must remain plausible enough that the procurement officer does not flag the bid distribution as anomalous, and the cover bidder cycles through cover roles across tenders generating cross-bid dispersion within firm (the *credible-cover-bidding* reading, R2; Marshall and Marx 2012; Asker 2010). The framework above is consistent with either reading. The within-stratum bridge of §6.2 discriminates between them: cobidders bid systematically closer to the eventual winner than FL non-cobidders ($d = -0.281$, $p < 10^{-6}$) with elevated within-firm dispersion ($d = +0.147$); a multivariate logit holding $\log(1 + \text{tenders_count})$ constant confirms both signs at $p < 10^{-3}$. The directions are opposite to R1 and consistent with R2; we adopt R2 as the operational reading. The framework's predictions about firm-level participation intensity (Proposition 2), deployment proximity to direct CADE defendants, and portfolio specialisation are unaffected by the R1/R2 distinction; only the bid-level reading of b_C is.

Appendix B. Core Supporting Tables and Figures

The sample-construction tables (modal mix, conditional descriptive statistics) and the heterogeneity tables (detection-regime quartiles, modal contrast, McCrary density, robustness to exclusion of CADE-involved items) referenced in the body are collected here, together with the always-loser participation distribution that motivates the median + 1.5×IQR cutoff (Figures B.1–B.2) and the detection-regime gradient (Figure B.3).

Appendix C. Identification Audits

Table C.1 reports the Cinelli and Hazlett (2020) robustness value and Oster (2019) bounds for the broad-sample association; the leakage decomposition and the permutation null behind the conservative benchmark are reported in Tables 8 and 7 of §6 (kept there as the discrimination centerpiece rather than as appendix audits). The leave-one-out

Table B.1: Tender Counts by Procurement Modality and Year

Year	Pregão	Convite	Total
2009	137,424	232,765	370,189
2010	134,181	253,196	387,377
2011	72,894	140,305	213,199
2012	129,664	305,808	435,472
2013	125,324	328,616	453,940
2014	118,274	310,947	429,221
2015	92,324	226,503	318,827
2016	57,804	122,841	180,645
2017	120,194	261,205	381,399
2018	60,518	121,698	182,216
2019	11,857	26,976	38,833

Notes: Winner-level observations from BEC (São Paulo) 2009–2019. Pregão = electronic reverse auction; Convite = sealed-bid invitation.

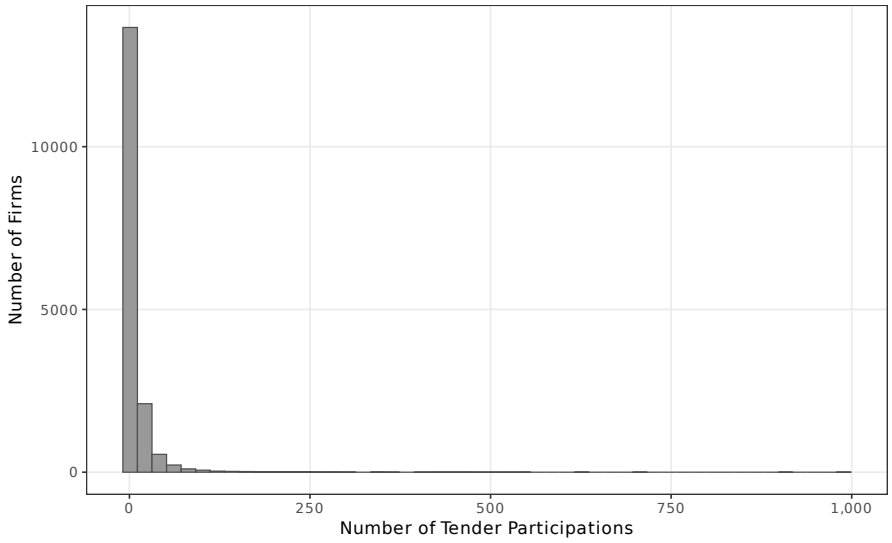

Figure B.1: Distribution of tender-item participation counts across always-loser firms in BEC, 2009–2019.

Table B.2: Conditional Descriptive Statistics: Residualized Prices by FL Status

	FL-Present (losers = 1)	FL-Absent (losers = 0)
Panel A: Unconditional		
Mean log(price)	4.9241	2.4938
Difference	2.4303	
Panel B: Conditional on Item + Year + PBU FE		
Mean residualized price	0.0472	-0.0024
SD residualized price	1.5734	0.8704
Difference (gap)	0.0495	
Overlap coefficient	0.978	
t-statistic	8.81	
p-value	0.000000	
Cohen's d	0.0390	
Observations	79,452	1,574,949

Notes: Residuals from regressing log(negotiated price) on item, year, and purchasing unit (PBU) fixed effects. The gap represents the mean difference in residualized prices between FL-present and FL-absent tenders, after absorbing common item-level, temporal, and PBU-level variation. Overlap coefficient = $2\Phi(-|\Delta\mu|/\sqrt{\sigma_1^2 + \sigma_0^2})$ measures the proportion of the two residual distributions that overlap; values near 1 indicate near-identical distributions. Cohen's d measures the standardized effect size.

Table B.3: Frequent-Loser Price Coefficient by Procuring-Unit Size and Procedure

Sub-sample	Coefficient	SE	N
By procuring-unit size quartile:			
Q1 (smallest buyers)	0.2138	(0.1670)	7,503
Q2	0.0111	(0.0513)	87,571
Q3	0.0660**	(0.0304)	302,258
Q4 (largest buyers)	0.0165	(0.0160)	1,257,069
By procedure type:			
Pregão (reverse electronic auction)	0.0933***	(0.0255)	546,549
Convite (sealed-bid invitation)	0.0382**	(0.0186)	1,107,852

Notes: Quartile gradient consistent with variation in the screening signal across detection regimes (§7.3); we do not identify a causal channel. Item and year FE (PBU FE for procedure split). SE clustered at item level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

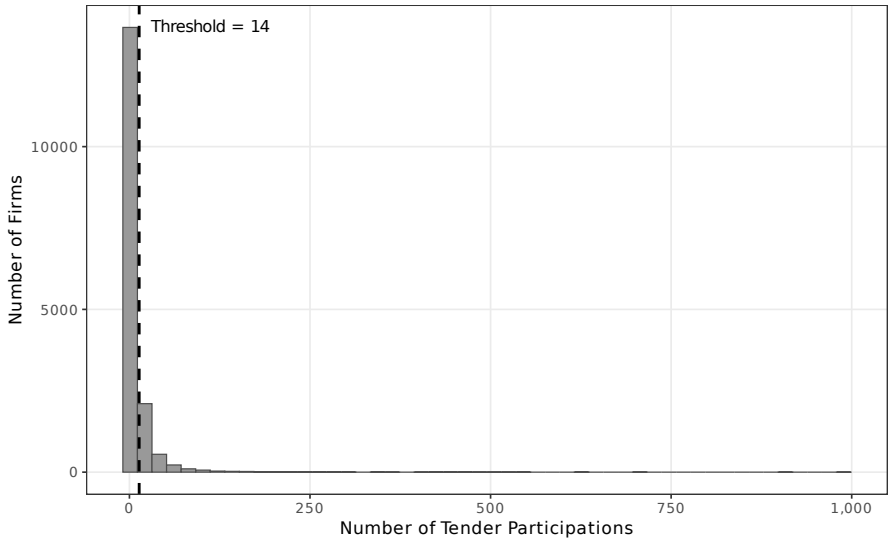

Figure B.2: Identification of the frequent-loser threshold via median + 1.5× IQR on the always-loser participation distribution; the cutoff yields 14 tender-items (13.5 statistical threshold) and flags 2,735 firms.

Table B.4: Frequent-Loser Price Coefficient by Modality (Pregão vs. Convite)

Specification	Pregão-only	Convite-only	Full sample	Ratio (P/C)
Binary FL14	$+0.0959^{***}$	$+0.0392^{**}$	$+0.0653^{***}$	2.45×
	(0.0256)	(0.0188)	(0.0216)	
	$p < 10^{-3}$	$p = 0.037$	$p = 0.003$	
Continuous log(1+ max tc)	$+0.0262^{***}$	$+0.0124^{**}$	—	2.11×
	(0.0063)	(0.0049)	—	
	$p < 10^{-4}$	$p = 0.011$	—	
Joint specification: both regressors				
Binary FL14 (with continuous control)	-0.0652	-0.0828^{**}	—	—
	(0.0449)	(0.0350)	—	—
Continuous (with binary control)	$+0.0399^{***}$	$+0.0305^{***}$	—	—
	(0.0118)	(0.0101)	—	—
Items	543,752	1,105,852	1,653,658	—
Item, year, PBU FE	YES	YES	YES	—
Cluster (item)	YES	YES	YES	—
Minimum-bidder rule binds	NO	YES	MIXED	—

Notes: Outcome is $\log p_{\text{negotiated}}$. The pregão coefficient is roughly 2.45× the convite coefficient on the binary specification. Under the screening framing (§5.1), we read the modal contrast as variation in the observability regime within which the screening signal is recovered, not as a positive test of any specific institutional channel. SEs clustered at item level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.5: McCrary Density Diagnostic at the Convite/Pregão Statutory Threshold

	Density at threshold			Modality
Threshold (Lei 8.666 art. 23)	Left	Right	Log-density disc.	Convite share gap
R$80,000 (pre-2018)	381	406	−0.063	0.098
R$176,000 (post-2017)	216	277	−0.250	0.092

Notes: Densities estimated by local linear smoothing of contract-record counts in ±50% windows around each statutory cap. The convite-share gap is the difference between the share of contracts using the convite modality just below versus just above the threshold. Both thresholds show small log-density discontinuities ($|\cdot| < 0.25$); neither shows the kind of bunching that would indicate strategic gaming of the modality assignment by buyers. The ~ 9.8 percentage-point convite-share drop across the cap is consistent with rule-driven assignment of value to modality, not with strategic selection.

Table B.6: Price Regressions Excluding CADE-Involved Markets

	(1) General+PBU	(2) Pregão
FL presence	0.0616***	0.0901***
	(0.0213)	(0.0256)
Observations	1,622,954	534,956
CADE tenders dropped	12,514	

Notes: All tenders involving any CADE-convicted firm are excluded. SE clustered at item level. *** p<0.01, ** p<0.05, * p<0.1.

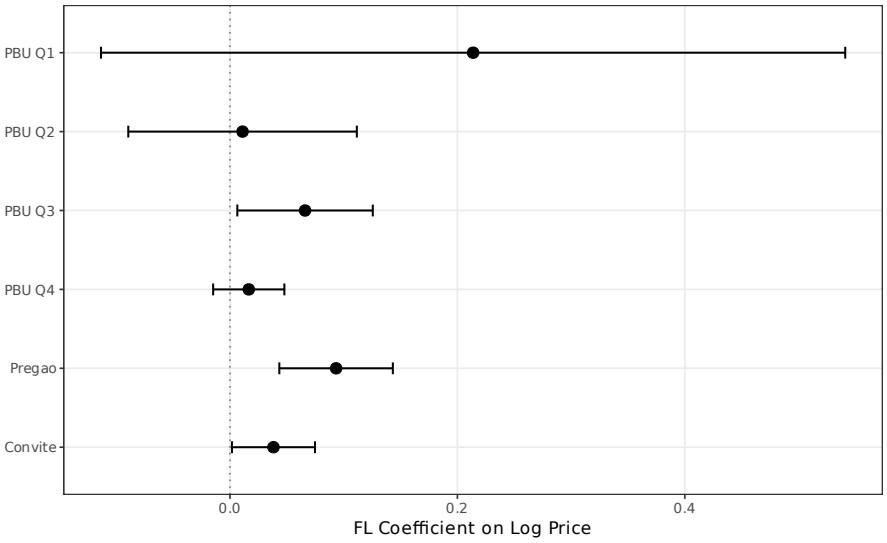

Figure B.3: Frequent-loser price coefficient by procuring-unit-size quartile; the Q1-vs-Q4 contrast (smallest to largest buyers) is 12.6×, intermediate quartiles imprecisely estimated. Companion to Table B.3.

instrumental-variable placebo (Table C.2) confirms the participation primitive operates through the threshold-crossing population the construct identifies; random subsamples of below-threshold always-losers do not produce comparable discrimination. Table C.3 reports the rolling-origin temporal-holdout AUC by test year (year-by-year decomposition of Figure 1 in the body). Table C.4 reports minimum-detectable-effect calculations for the failed RDD and DiD designs of §5.1: observed-to-MDE ratios well below one indicate the null findings are structurally underpowered, not refutations of the underlying mechanism.

Table C.1: Oster (2019) Coefficient Stability Bounds

	Full sample	Pregão	Convite
$\hat{\beta}_{\text{short}}$ (item + year FE)	0.0676	0.0954	0.0507
$\hat{\beta}_{\text{full}}$ (+ PBU FE)	0.0636	0.0933	0.0382
$\tilde{R}^2_{\text{short}}$	0.8790	0.8737	0.8857
$\tilde{R}^2_{\text{full}}$	0.8860	0.8821	0.8932
$R^2_{\max} = \min(1, 1.3\tilde{R}^2)$	1.0000	1.0000	1.0000
$\hat{\delta}$ (Oster bound)	261.6	634.00	43.62
$\beta^*(\delta = 1)$	-0.0005	0.0643	-0.1395
Observations	1,654,401	546,549	1,107,852

Notes: $\hat{\delta}$ is the degree of selection on unobservables (relative to observables) required to drive $\hat{\beta}$ to zero (Oster, 2019). $|\hat{\delta}| > 1$ indicates robustness: unobservables would need to be more important than observables to explain the coefficient. $\beta^*(\delta = 1)$ is the bias-adjusted coefficient assuming equal selection. $R^2_{\max} = 1.3\tilde{R}^2$ following Oster's recommended bound. Short model: item + year FE. Full model adds PBU FE (and convite control for full sample).

Appendix D. Threshold and Specification Robustness

The IQR-multiplier sensitivity (Table D.1) places the headline price coefficient as positive and significant under five multipliers ranging from 1.0× to 3.0× IQR, declining monotonically as the cutoff tightens—the pattern the framework's continuous primitive predicts (§8.1). Standard-error clustering (Table D.2) under item-level (baseline), procuring-unit-level, and two-way Cameron et al. (2011) clustering keeps the coefficient significant at $p < 0.01$ under all three regimes.

Table C.2: Placebo IV: Sub-Threshold Always-Losers as Instrument

	First Stage	2SLS
	FL presence	Log Price
Sub-threshold supply (LOO)	0.000040***	
	(0.000007)	
F-statistic	33.0	
FL presence (instrumented)		5.2445***
		(1.3200)

Notes: Placebo instrument uses supply of sub-threshold always-losers (below the IQR cutoff). The sub-threshold first stage ($F = 33$) exceeds the Stock and Yogo (2005) 10% maximal IV size critical value of 16.38 for a single instrument. The implausibly large 2SLS coefficient (5.24) is therefore not attributable to weak-instrument bias. Rather, it reflects the absence of a genuine cover-bidding mechanism: sub-threshold always-losers do not drive procurement price anomalies, confirming that the main instrument's price effect operates through the cover-bidding channel rather than through correlated demand shocks. SE clustered at item level.

Table C.3: Temporal-Holdout AUC by Test Year

Test year	AUC	95% CI	N firms	N CADE
2014	0.819	[0.779, 0.860]	8,631	88
2015	0.817	[0.776, 0.858]	10,122	102
2016	0.851	[0.821, 0.881]	11,699	127
2017	0.862	[0.832, 0.891]	13,057	148
2018	0.897	[0.877, 0.918]	14,391	161
2019	0.922	[0.912, 0.933]	15,597	181

Notes: Each row trains the construct on years preceding the test year (rolling-origin scheme starting in 2009) and evaluates AUC against the cobidder ground truth realized in the test year. Bootstrap 95% CI from 1,000 resamples. Source: `scripts/17_temporal_holdout_roc.R`.

Table C.4: Minimum Detectable Effects for Failed RDD and DiD Designs

Test	Observed coef	SE	MDE @ 80%	MDE @ 50%	N	Obs/MDE
RDD R$80k pre-Decreto, FL prevalence	−0.0038	0.0033	0.0092	0.0065	7,836	0.41
RDD R$176k post-Decreto, FL prevalence	+0.0054	0.0170	0.0476	0.0333	1,053	0.11
RDD R$80k pre-Decreto, convite share (1st-stage)	+0.0046	0.0198	0.0553	0.0387	5,324	0.08
RDD R$176k post-Decreto, convite share (1st-stage)	+0.1556	0.0759	0.2126	0.1488	472	0.73
DiD Decreto, FL prevalence	−0.0059	0.0105	0.0294	0.0206	3,676	0.20
DiD Decreto, log(value) — placebo	+0.0616	0.0574	0.1608	0.1125	3,676	0.38
DiD Decreto, n_firms	−0.0022	0.0016	0.0044	0.0031	3,676	0.49
DiD Decreto, convite share (1st-stage)	+0.0234	0.0356	0.0998	0.0698	3,676	0.23

Notes: Power calculations for the design-based identification strategies that returned null. The MDE columns report t effect detectable at 80% and 50% power for the realized N. Observed-to-MDE ratios well below one indicate the null is underpowered, not a refutation of the underlying mechanism. Source: `scripts/23_mde_calculations.R`.

Table D.1: Price Coefficient Across IQR Multiplier Thresholds

Multiplier	Threshold	FL Firms	Coef.	SE	Observations
1.0 × IQR	10	3,442	0.0791***	(0.0227)	1,654,401
1.5 × IQR	14	2,735	0.0636***	(0.0215)	1,654,401
2.0 × IQR	17	2,093	0.0598***	(0.0221)	1,654,401
2.5 × IQR	20	1,778	0.0543**	(0.0228)	1,654,401
3.0 × IQR	24	1,456	0.0501**	(0.0237)	1,654,401

Notes: DV: log negotiated price. Each row re-estimates the baseline specification (item + year + PBU fixed effects) using a different IQR multiplier to define the FL threshold. The threshold equals median $+k\times$ IQR of the tenders-count distribution among always-losers. The baseline (1.5× IQR) is highlighted. SE clustered at item level. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table D.2: Alternative Clustering of Standard Errors

Clustering level	Coef.	SE	95% CI
Item level (baseline)	0.0636***	(0.0215)	[0.0214, 0.1058]
PBU level	0.0636***	(0.0144)	[0.0353, 0.0919]
Two-way (item + PBU)	0.0636***	(0.0240)	[0.0165, 0.1107]

Notes: DV: log negotiated price. All rows report the same baseline specification (item + year + PBU fixed effects) with different clustering of standard errors. The point estimate is invariant to clustering; only the standard errors and confidence intervals change. The two-way cluster uses the Cameron et al. (2011) multi-way approach. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Appendix E. Generalisation Audit and Detector Comparison

The operational-metrics temporal-holdout audit (Table E.1) referenced in §9.3 and the same-sample incremental-AUC comparison (Table E.2) referenced in §10 are reported here, together with the cross-sector external-validity scope (Table E.3) that quantifies the within-product-code identifying variation underwriting the headline price gap and the AUC decomposition (Table E.4) that reports the marginal contribution of each feature block to discrimination on the cobidder ground truth. The participation primitive $\log(1+\text{tenders_count})$ alone reaches AUC near the full-feature ceiling; bid-based features contribute non-trivially and dropping the participation primitive collapses discrimination to bid-distribution-only performance, supporting the framework's identification of the participation primitive as the dominant signal at the award layer.

Table E.1: Operational Deployment Metrics: Precision, Recall, and Lift at Top-k

	In-sample (full 2009–2019)				Temporal holdout (score: 2009–2016)			
Top-k	TP	Precision	Recall	Lift	TP	Precision	Recall	Lift
50	15	0.300	0.078	26.2×	1	0.020	0.005	1.7×
100	17	0.170	0.088	14.8×	7	0.070	0.036	6.1×
250	40	0.160	0.207	14.0×	19	0.076	0.098	6.6×
500	66	0.132	0.342	11.5×	35	0.070	0.181	6.1×
1,000	97	0.097	0.503	8.5×	66	0.066	0.342	5.8×
Reference quantities								
Always-loser pool size N					16,843			
Cobidder positives N_+					193			
Base cobidder rate					0.0115			
ROC AUC (log(1+tc))		In-sample: 0.939				Temporal holdout: 0.864		

Notes: Score: $\log(1 + \text{tenders_count})$, ranked descending across the always-loser pool. Lift is precision divided by the base cobidder rate. The in-sample columns score firms using 2009–2019 participation counts; the temporal-holdout columns score using only 2009–2016 participation, evaluated against the same 193 cobidder labels. The temporal-holdout numbers are the generalization reference for the screening statistic; the in-sample columns over-state precision by approximately 50% at top-500 and are reported for transparency. The screening statistic requires only contract-award records (winner identity, participant identity, item identifier); no bid microdata.

Table E.2: Incremental Value of Loser-Side Concentration Relative to the Full Imhof Benchmark

Model	Data requirements	AUC	95% CI	Δ vs Imhof full	DeLong p
Imhof full pipeline	All bid values within each tender	0.846	$[0.819, 0.873]$	+0.000	—
Binary FL flag	Winner + participants only	0.881	$[0.871, 0.892]$	+0.035	0.014
Continuous participation count	Winner + participants only	0.877	$[0.857, 0.898]$	+0.031	0.077
Imhof full + binary FL	Bid microdata + award records	0.942	$[0.927, 0.957]$	+0.096	< 0.001
Imhof full + participation count	Bid microdata + award records	0.944	$[0.929, 0.958]$	+0.098	< 0.001

Notes: All models are evaluated on the exact same always-loser subsample for which the full bid-distribution feature set is available. The relevant comparison is therefore not "does FL beat Imhof everywhere," but whether participation-only information remains informative when rich bid microdata are absent and whether it adds non-redundant signal when those microdata are present. The combined specifications answer the second question directly.

Table E.3: External Validity Scope: Modality and Commodity-Service Boundaries

Dimension	Group	Metric	Value	95% CI
coverage	Commodity	share_items	0.887	—
coverage	Service	share_items	0.113	—
dominant_item_class	Commodity	auc_log_tc	0.905	$[0.879, 0.931]$
dominant_item_class	Service	auc_log_tc	0.966	$[0.961, 0.971]$
item_class_price	Commodity	coef_losers	0.045	$[0.002, 0.088]$
item_class_price	Service	coef_losers	0.128	$[0.024, 0.231]$
modal_primary_auc	convite_primary	auc_log_tc	0.816	$[0.758, 0.874]$
modal_primary_auc	pregao_primary	auc_log_tc	0.952	$[0.946, 0.958]$

Notes: Item classes use a transparent text rule on BEC item descriptions. Rows labelled `modal_primary_auc` reuse the always-loser firm split by primary modality; rows labelled `item_class_price` report the within-item price association separately for commodity-like and service-like items. The point is not to claim exportability to all procurement, but to document that the paper speaks most naturally to standardized-goods environments and selected simple services.
Source: `scripts/52_external_validity_scope.R`.

Table E.4: AUC Decomposition Across Feature Sets

Model	Features	AUC	95% CI	Marginal AUC vs. A
A	is_fl + imhof_cv + imhof_spread + tenders_count + n_bids	0.939	[0.928, 0.949]	0.939
B	imhof_cv + imhof_spread + tenders_count + n_bids	0.936	[0.924, 0.948]	0.003
C	is_fl	0.887	[0.868, 0.905]	0.052
D	imhof_cv + imhof_spread	0.785	[0.754, 0.816]	0.154

Notes: Random-forest classifiers trained on the cobidder ground truth, with five feature sets. Model A is the full set; subsequent models drop one block. The marginal contribution column reports the AUC delta from removing the listed feature(s) relative to A. Source: `scripts/20_auc_decomposition.R`.

Appendix F. Adversarial-Adaptation Simulation

The screen's behaviour under deliberate adversarial adaptation (§11.2) is bounded but not invariant. We examine four strategic adaptations. Each holds the screening statistic fixed and modifies the always-loser pool to mimic plausible evasive behaviour, and we recompute the firm-level AUC of $\log(1 + \text{tenders_count})$ against the 193 CADE-cobidder labels under each (Table F.1). Two are observable adaptations the screen absorbs cleanly. Rotation (20% of cover bidders retire annually and are replaced by fresh CNPJs) and occasional wins (5% of threshold-crossing firms win one tender) leave AUC essentially unchanged: the bright-line wins = 0 filter eliminates the small share of newly-winning cover bidders—a feature of the framework's separating equilibrium (Lemma 1), not a heuristic robustness—and rotation does not compress the $\log(1 + \text{tenders_count})$ ranking statistic across the always-loser pool.

Two are threshold-aware adaptations that degrade discrimination. CNPJ splitting (each frequent loser splits into two CNPJs carrying half the participation count) degrades AUC by 3.7 percentage points; threshold-aware capping (firms cap participation at 13 tender-items to stay below the cutoff) degrades AUC by 12.7 points; the combined attack (capping plus 20% rotation) drives AUC from 0.928 to 0.632, a 29.5-point compression. These adaptations target the binary cutoff specifically; the continuous primitive $\log(1 + \text{tenders_count})$ that Proposition 2 identifies as the sufficient ranking statistic is not directly altered, but the binary discretisation is. Threshold capping is the most exposed margin: a cartel that knows the cutoff can instruct cover bidders to remain below it.

The implication for the screening-vs-forensic-stage architecture of §10 is asymmetric. Under combined adaptation AUC drops to 0.632—still above random matching, still

Table F.1: Adversarial-Adaptation Simulation: AUC Degradation Under Strategic Cartel Adaptations

Scenario	AUC	Δ vs baseline	Retention
Baseline (no adaptation)	0.928	+0.000	100.0%
Rotation: 20% of cover bidders retire and are replaced	0.931	+0.003	100.3%
Occasional wins: 5% of FL14 firms get one win	0.929	+0.002	100.2%
CNPJ splitting: each FL14 firm splits into two CNPJs	0.891	−0.037	96.0%
Threshold cap: firms cap participation at 13 tender-items	0.801	−0.127	86.3%
Combined: threshold cap + 20% rotation	0.632	−0.295	68.2%

Notes: Each scenario applies a strategic adaptation to the BEC always-loser pool, holding the construct (log(1+tenders_count) ranking against 193 CADE cobidder labels) fixed. The construct is resilient to rotation and to occasional wins (the bright-line filter eliminates the small share of newly-winning cover bidders without compressing the ranking), and degrades modestly under CNPJ splitting. It is substantially vulnerable to threshold-aware capping and severely degraded when capping is combined with rotation: a sophisticated cartel that targets the threshold can compress AUC by roughly 30 points. Retention is the fraction of baseline AUC preserved.

informative as a screening statistic, but substantially reduced. The forensic stage is unaffected by threshold-aware adaptation at the screening layer: bid-distribution moments operate on a different data envelope and are robust to the participation-pattern engineering that defeats the binary rule. A combined two-stage architecture in which a refreshable screening rule feeds a forensic stage on bid microdata is therefore more adaptation-robust than either component alone—the empirical evidence in §10 shows the same conclusion by distinct means. The screening statistic is not adaptation-proof; practical lifetime depends on whether the threshold can be periodically refreshed against new data and whether the screening stage is deployed alongside the bid-layer forensic stage where microdata are available.

Appendix G. Staggered Designs Attempted and Rejected

Two design-based strategies that would extract a causal estimate return null at the available bandwidths (§11.1). The sharp-RDD failure (gated by the McCrary density test) is documented in Table B.5 of Online Appendix B; this appendix reports the parallel diagnostic for the difference-in-differences strategy exploiting Decreto 9.412/2018's 2018 raise of the convite cap. The Sun and Abraham (2021) interaction-weighted estimator yields imprecise coefficients with non-monotone dynamic patterns incompatible with

the smooth treatment-onset narrative the cap raise would imply (Table G.1); a stacked difference-in-differences design constructed around the cap-raise event window returns coefficients that flip sign across stacks and lack a clean pre-period (Table G.2). We treat both estimators as null at the available bandwidths. The interpretive consequences are limited to identification scope: the screening statistic does not require the cap raise to be a clean treatment, and the broad-sample β of §7.1 does not depend on the staggered design surviving. What the null does constrain is alternative framings of the empirical object that would require a causal identification of β; the screening framing trades that claim for the screening-value interpretation of β and reports the failures here for full disclosure.

Table G.1: Staggered DiD: Callaway & Sant'Anna (Restricted Sample)

	C&S (2021)		TWFE	
Outcome	ATT	SE	Pre-trend	Post
Log Price	0.0143	(0.0396)	-0.0131	-0.0079
Log Firms (excl. FL)	-0.0220	(0.0333)	-0.0088	0.0113
Panel B: Statistical Power				
MDE (5%, two-sided)	0.0776			
MDE (80% power)	0.1109			
OLS benchmark	0.064			
Panel D: Sharp Entry Subsample				
C&S ATT	0.0143	(0.0389)		
Sharp entry markets	1,511 treated, 18,266 control			
Markets	19,777 (1,511 treated, 18,266 control)			
Sample restriction	$\geq$3 pre & $\geq$3 post observations			

Notes: Panel A: C&S = Callaway & Sant'Anna (2021) doubly-robust estimator. Panel B: MDE = minimum detectable effect at stated significance/power levels. Panel C: Rambachan & Roth (2023) sensitivity bounds under smoothness restriction $\bar{M}$ on maximum change in the slope of the trend. Panel D: restricts treated markets to those with zero FL presence before treatment year.

Table G.2: Stacked DiD vs. Callaway & Sant'Anna

	Stacked (Cengiz et al.)		C&S (2021)	
Outcome	ATT	SE	ATT	SE
Log Price	-0.0059	(0.0138)	0.0143	(0.0405)
Log Firms (excl. FL)	0.0162*	(0.0084)	-0.0220	(0.0339)
Stacked obs.		715,116		
Cohorts		6		
Event window		±5 years		
Cohort × Market FE	YES		—	
Cohort × Year FE	YES		—	
Clustering		Market level		

Notes: Stacked regression following Cengiz et al. (2019). Each cohort-year sub-experiment includes treated markets entering in year g and all never-treated markets, within a symmetric ±5-year event window. Cohort-specific market and year fixed effects absorb level differences across sub-experiments. SE clustered at market level. C&S = Callaway & Sant'Anna (2021) doubly-robust estimator from Table G.1. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

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